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Mystery of the Disappearing Dogfish: Transboundary Analyses Reveal Steep Population Declines Across the Northeast Pacific With Little Evidence for Regional Redistribution

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ABSTRACT

Quantifying broad-scale population trends and distribution change is critical for effective management and conservation of marine species, particularly under climate change. However, fragmented regional survey data often hinder such efforts for transboundary populations. A prime example is Pacific Spiny Dogfish (*Squalus suckleyi*, Squalidae), a small shark with a remarkably slow life history and wide-ranging distribution. Dogfish are now caught incidentally but were heavily fished along the Pacific US–Canada coast ≈80 years ago. Reports on local population trends have conflicted along the coast, suggesting that movement between regions may be responsible. We fit spatiotemporal models integrating data from 10 surveys to synthesise trends in biomass, abundance, distribution and thermal niche for dogfish across their entire eastern Pacific Ocean range. Prior to 2003, Alaskan biomass increased through the 1990s whereas California to British Columbia indices were variable and imprecise. However, during 2003–2023, we found a coastwide 51% (95% CI: 38%–61%) decline in dogfish biomass with mature females and immature dogfish showing the largest proportional declines. Regionally, declines were steepest for the US West Coast (71%–85%) and Canada (58%–82%), while Alaska showed less severe declines (13%–54%). Off the US West Coast, dogfish shifted into deeper waters as temperatures in their habitat increased, but these patterns do not explain the coastwide declines. Our results suggest population declines are primarily driven by reduced abundance rather than between-region movement, indicating elevated coastwide conservation concern and helping focus investigations of causal mechanisms.

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1 | Introduction

Assessing broad-scale trends in population size and distribution is important for effective fisheries management and conservation, particularly as marine ecosystems respond to climate change. However, determining population trends can be challenging for wide-ranging, transboundary marine species (Mccauley et al. 2015; Kyne et al. 2020). Survey data for such species may be collected by multiple agencies and by distinct scientific surveys following separate protocols (Maureaud et al. 2021). Population assessments by government agencies tend to focus on regional data with limited communication with nearby agencies. While such regional stock assessments provide important information on local population size and trends, a broader perspective is important for transboundary species—particularly with climate change driving the redistribution of species across international borders and scientific surveys (Maureaud et al. 2021; O'Leary et al. 2021).

The Pacific Spiny Dogfish (*Squalus suckleyi*, Squalidae; hereafter 'dogfish') is an example of a marine species with a wide-ranging, transboundary distribution. In the eastern North Pacific, dogfish inhabit waters from Baja California to the Bering Sea (Ketchen 1986) and are commonly found from Oregon to the Gulf of Alaska (Ketchen 1986; Orlov et al. 2012; Gasper and Kruse 2013). Inference on population trends for dogfish is limited as most scientific surveys have neither the temporal nor spatial coverage to match their slow life history, long lifespan (maximum age around 80–85 years), or their wide-ranging distribution (Mcfarlane and King 2009; Tribuzio and Kruse 2012; Taylor et al. 2013; Gasper and Kruse 2013). Stock assessments estimate regional population trends from local surveys. However, it is unclear if trends in one region represent the population as a whole or whether local trends could be due to movement to neighbouring regions or changes in frequency of occurrence by size and sex due to their complex size- and sex-specific movements and aggregations.

While dogfish populations have previously been considered stable and their extinction risk low across the eastern North Pacific coast (Bigman 2013; Galluci et al. 2011), there is evidence that regional differences underlie this broader assumed stability. In the Canadian portion of their range, uncertainty regarding the abundance of mature females, combined with the species' biological characteristics such as low fecundity, long generation time and gestation period and vulnerability to overfishing led to an extinction risk categorization of 'Special Concern' (COSEWIC 2011). More recently, a stock assessment for dogfish in British Columbia, Canada (BC) estimated that by the year 2000, spawning output of dogfish was between 10% and 30% of unfished levels and total biomass at ≈30% of unfished levels (DFO 2025; Anderson et al. 2025). Since then, steep declines in dogfish biomass were observed in fisheries-independent survey indices in BC (Anderson et al. 2019; Anderson and English 2022) and spawning output was estimated to be 6%–12% of unfished levels by 2024 and under the biological limit reference point with high (>0.99) probability (DFO 2025; Anderson et al. 2025). On the United States (US) West Coast (the outside waters of Washington, Oregon and California), the dogfish population was estimated to be relatively stable in recent years; however, the population was estimated to be at 34% of unfished levels by 2021 and thus above overfished thresholds but below management targets (Gertseva et al. 2021). On the other hand, dogfish population trends in Alaska are reported to be stable or increasing based on commercial and survey catch rates (Conrath and Foy 2009; Tribuzio and Kruse 2011; Tribuzio et al. 2022). Most changes in regional population size have been suggested to be driven by climate-related movements of individuals outside of each region (Conrath and Foy 2009; Orlov et al. 2012; Gertseva et al. 2021; Shiffman et al. 2022).

Here we use novel spatiotemporal modelling methods to combine fishery-independent surveys spanning multiple jurisdictions and overcome previous analysis limitations and provide insight on regional and coastwide population trends, distribution and hotspots of change (e.g., Maureaud et al. 2021; O'Leary et al. 2021, 2022). We ask (1) What is the California–Gulf of Alaska coastwide population trend for dogfish over the last two decades? (2) How consistent are these trends across regions and might region-specific trends reflect a climate-driven range shift? (3) How consistent are these trends across maturity groups and sexes? (4) What does the distribution of dogfish by maturity and sex look like along the eastern North Pacific coast and where are hotspots of population density change? (5) Is the depth and temperature distribution of dogfish stable through time?

Our findings reveal declines along most of the coast, with particularly steep declines in British Columbia and the US West Coast. We found declines across all maturity groups and sexes, with the steepest declines in immature and mature female dogfish. Hotspots of decline occurred in southeastern Alaska, off southern Vancouver Island and northern Washington State. We also observed a general deepening of dogfish density off British Columbia and the US West Coast, but not in Alaska, potentially suggesting tracking of preferred temperatures in the US West Coast and BC regions.

2 | Methods

Broadly, our analytical approach used spatiotemporal models to standardise dogfish fisheries independent survey data in the eastern North Pacific. We used model predictions to summarise coastwide and regional population trends, population trends by maturity and sex categories, spatially varying local population trends, and to summarise changes to depth distributions. The Supporting Information provides a detailed description of data and analytical steps; here we provide a summary of key methodologies.

2.1 | Data

We collated data from ten bottom trawl surveys ranging in latitude from California to the Gulf of Alaska that, when considered together, spanned 1980 through to 2023. These data are from three regions: the outside waters of Washington, California and Oregon (US West Coast) (Keller et al. 2017; Wetzel et al. 2024), British Columbia (BC), Canada (Sinclair et al. 2003; Anderson et al. 2019) and the Gulf of Alaska (GOA), US (NOAA Fisheries 2022; Stauffer 2004). All trawl surveys were conducted with broadly similar gear and collection protocols; however, we

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tested for differences in survey catchability within our models (Figures S12 and S13). We obtained longline survey data from the International Pacific Halibut Commission (IPHC) setline survey (IPHC 2022, 2024). These data span 1998–2022, although the survey has not covered the full survey domain in all years (Figure S8).

To model trends in population density and distribution by sex and maturity classes, we split set-level catch weight. Catch weight was split by the proportion of weight from sub-sampled individuals that belonged to each sex and maturity class (immature, maturing, mature). Individuals were assigned to groups using lengths at different probabilities of maturity thresholds (Supporting Information). As the IPHC survey has only collected dogfish lengths since 2011, and we were interested in patterns that began in the early 2000s, we focused on the trawl surveys when examining patterns by maturity. In all regions, spatial coverage and collection of lengths have been relatively consistent for trawl surveys since 2003.

2.2 | Model Structure

To combine fisheries-independent survey data across the eastern North Pacific region of interest, we fit geostatistical generalised linear mixed effects spatiotemporal models using the R package sdmTMB (Anderson et al. 2024). Our full models included quadratic effects for bottom depth, catchability effects, Gaussian Markov random fields (GMRFs) (Rue and Held 2005; Lindgren et al. 2011) accounting for latent spatial effects, and GMRFs structured as random walks to account for latent spatiotemporal effects (Supporting Information). This model form—although often with independent year means and independent spatiotemporal GMRFs—is a common form of model applied to fish survey data (e.g., Shelton et al. 2014; Thorson et al. 2015; Thorson 2019; Anderson et al. 2024; Thorson and Kristensen 2024). Given that many of these surveys were conducted biennially or triennially, we used a random walk on the spatiotemporal effects to allow estimation in areas that were not surveyed in some years and to avoid attributing spatial sampling variability to temporal variability. A similar approach was taken in Anderson and English (2022), DFO (2024) and Webster et al. (2020) with the latter using a related first-order autoregressive process.

The coastwide model includes the United States and Canada Pacific coasts (Alaska, British Columbia, Washington, Oregon and California). The coastwide trawl model spanned 2003–2023 while the coastwide IPHC model spanned 1998–2022. The regional models included (1) Gulf of Alaska, (2) British Columbia, Canada and (3) Washington, Oregon, California with variable years for trawl (1993–2021, 1984–2023 and 1980–2023 for each region, respectively). The trawl survey catch weight data were fit with Poisson-linked delta-models (Thorson 2018), with a Bernoulli likelihood for encounter/non-encounter of dogfish, and a (mixture) lognormal likelihood (Thorson et al. 2011) for positive density (accounting for area swept by the trawl net). For most of these models, residual diagnostics (Dunn and Smyth 1996; Smith 1985; Hartig 2022) indicated that a mixture approach was necessary to account for the occasional extreme catch events of dogfish in trawl surveys (Thorson et al. 2011). We specified a 1% probability that a given positive catch density was drawn from an 'extreme' catch distribution—a lognormal distribution with a higher mean but the same dispersion parameter. We fit models to the IPHC count data (as captured dogfish catch weight is not recorded) with a negative binomial (NB2; Hilbe 2011) likelihood and a log link with an offset accounting for hooks observed. To determine hotspots of biomass density through time, we fit coastwide models with a spatially varying coefficient of year constrained as a GMRF (Hastie and Tibshirani 1993; Barnett et al. 2021). Some regional and maturity-specific models included fewer encounters and/or less consistent spatial coverage than the full dataset and therefore required simplified model structures (Supporting Information).

2.3 | Parameter Estimation

We fit all models with the sdmTMB (Anderson et al. 2024) R (R Core Team 2024) package, which employs the Stochastic Partial Differential Equation (SPDE) approach (Lindgren et al. 2011) to approximate Gaussian random fields with GMRFs for computational efficiency. sdmTMB uses the fmesher (Lindgren 2023) R package to generate input matrices to carry out SPDE precision matrix calculations, and calculates the marginal log likelihood by integrating over random effects with the Laplace approximation as implemented in the TMB (Kristensen et al. 2016) R package. The negative marginal log likelihood is then minimised via a non-linear minimizer and standard errors on derived quantities of interest are calculated via a generalised delta method (Kristensen et al. 2016).

We conducted a variety of model diagnostics, model comparisons and assessments of model convergence (Supporting Information). We compared model structures via the marginal AIC (Akaike information criterion; Akaike 1973) and assessed simulation-based residual diagnostics (Dunn and Smyth 1996; Smith 1985; Hartig 2022, Supporting Information) to select a model structure. To assess whether our coastwide models generated reasonable representations of the regions, we compared predictions from the coastwide model against those from models fit independently to the regional data. These regional models also provided longer time series, but with more limited spatial coverage and variability in survey timing.

2.4 | Calculating Population Indices and Density-Weighted Depth and Temperature

To visualise spatial predictions, calculate area-weighted indices of abundance, and calculate biomass-weighted mean depth occupied, we required a discretized grid covering the domain of the surveys. This allowed for Monte Carlo integration by expanding the variable of interest to each grid cell and calculating a sum across the grid cells (Thorson and Kristensen 2024). We therefore developed 4 km × 4 km and 3 km × 3 km grids that covered the survey domains of the coastwide trawl and longline survey data, respectively (Supporting Information).

We characterised trends in the trawl and IPHC survey indices two ways. First, for the trawl survey indices of biomass, we fit generalised linear models (GLMs) with log links and gamma

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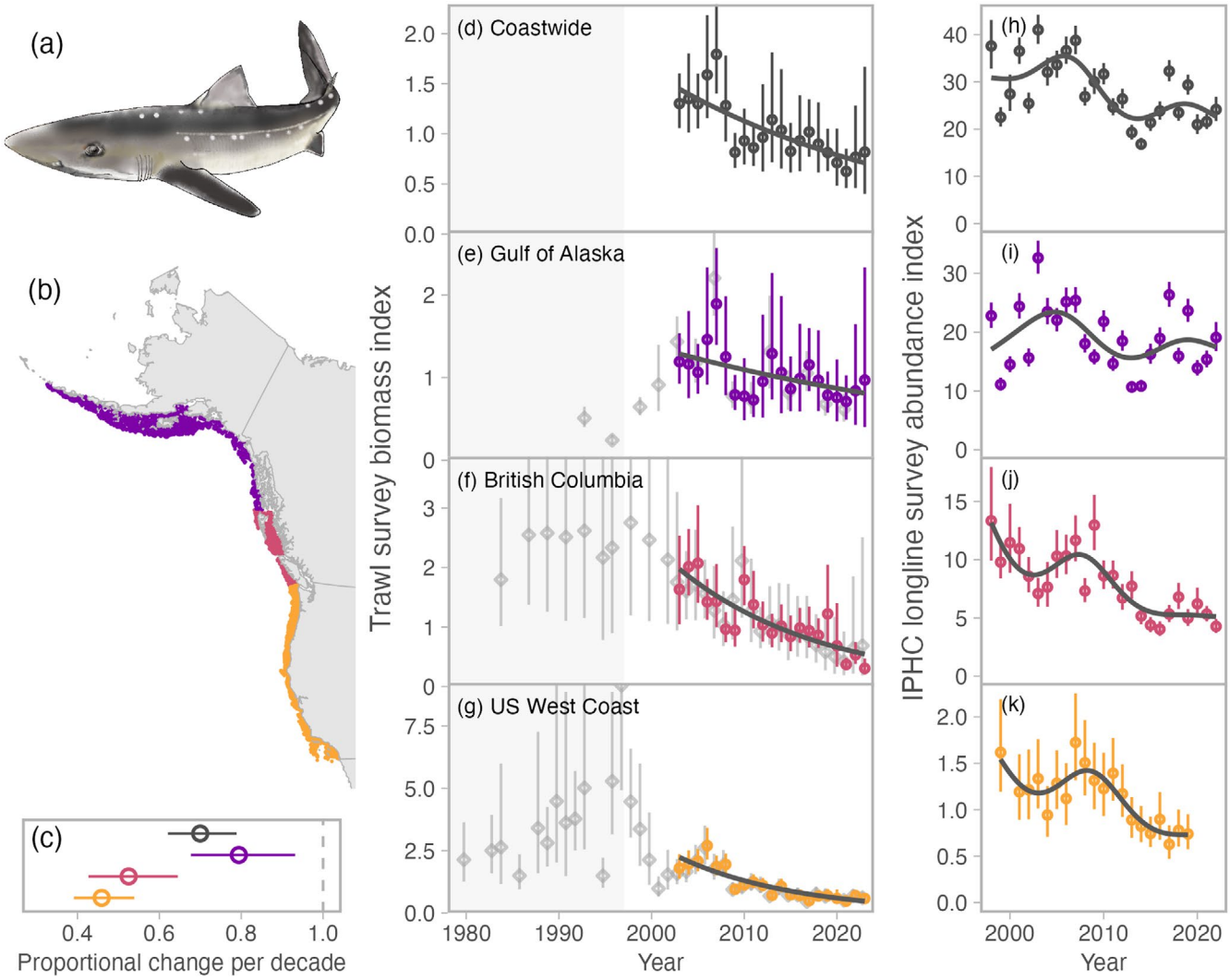


FIGURE 1 | Trends in population density of dogfish across the eastern North Pacific coast from the Gulf of Alaska to California. (a) Drawing of a Pacific Spiny Dogfish. (b) Distribution of fisheries independent trawl survey data and regions of interest. (c) Slope coefficients describing the multiplicative population change per decade from 2003 onwards for the trawl surveys. (d–g) Standardised and normalised population trends coastwide and by region from trawl surveys. Grey diamonds represent estimates from regional models; coloured circles represent estimates from a coastwide model. Fitted lines represent log-linked generalised linear models with gamma error fit to the coastwide model trends from 2003 onwards (slopes shown in panel c). All time series are normalised to have a geometric mean of 1 for 2003–2022. Grey shading indicates the time period prior to the start date of the IPHC longline survey. (h–k) Standardised relative abundance from the IPHC longline survey for the same regions as for trawl. Fitted lines represent a log-linked generalised additive model with gamma error. Throughout, dots and line segments represent means and 95% confidence intervals, respectively. Dogfish drawing created by Edith Kraus and used with permission.

error to the mean index values by year given the curvilinear declines. We characterised trends for the longline indices using generalised additive models (GAMs) fit with the mgcv R package (Wood 2017), again with log links and gamma error. While we report declines using a log-linear relationship for trawl indices, we acknowledge fluctuations around that trend that mirror patterns in IPHC trends.

We calculated area-weighted indices of relative biomass and abundance by predicting from our spatiotemporal models on the survey grids and summing the product of population density and cell area across the domains. We calculated biomass-weighted mean depth by predicting biomass density for each cell, multiplying the density by cell area to expand to biomass, and taking the biomass-weighted mean depth across grid cells.

3 | Results

3.1 | Overall Population Index Trends

We observed coastwide declines since the beginning of the coastal time series (2003–2023, Figure 1d) but with variation in magnitude by region (Figure 1e–h). We observed curvilinear declines in dogfish biomass along the entire coast since 2003 (Figure 1c), and GLMs assuming exponential declines characterised the average proportional change per decade (Figure 1c). During this time period, the model indicated an average 51% (95% CI: 38%–61%) decline (Figure 1c). Regional declines on the US West Coast (79%; 95% CI: 71%–85%) and in BC (72%; 95% CI: 58%–82%) during 2003–2023 were similar but lower in the Gulf of Alaska (37%; 95% CI: 13%–54%) (Figure 1). The biomass

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indices generated at the regional level but using the coastwide spatiotemporal model were roughly equivalent to those resulting from regional models but with the added advantage of reducing uncertainty and allowing for the estimation of coastwide metrics of interest (Figure 1e–g).

Prior to our coastwide trawl model beginning in 2003, regional data suggested varying patterns in biomass indices. In the Gulf of Alaska, surveyed biomass was lower in the 1990s before peaking in 2007, representing a 330% increase in dogfish biomass during this time (Figure 1e). Dogfish biomass has since declined from this peak, and as of 2021, biomass was estimated to be 20% greater than the earliest estimate in 1993 (Figure 1e). In BC, the biomass index prior to 2003 was generated from a survey conducted in a small geographic area of BC's coast (Figure S4). When predicted across BC's coast, this index estimated biomass in BC to be higher than predictions in 2003; however, there is considerable uncertainty since predictions were made across a much larger area than the survey domain (Figure 1f). Estimated biomass trends prior to 2003 on the US West Coast were variable and highly uncertain owing to differences in early surveys, but higher than current estimates (Figure 1g)—fluctuations occurred from lows in the early 1980s and 1995 to the highest estimate in 1997. Except for 2006, every biomass estimate from 2001 onwards is estimated to be below the long-term biomass average (Figure 1g).

Similar to data from trawl surveys, data from the IPHC longline survey also suggest coastwide declines in abundance, albeit less steep (~36% decline 1998–2022). Declines were also apparent in BC and on the US West Coast with non-linear patterns through time (Figure 1h–k). The trends across the regions were fairly consistent in their timing, except the Gulf of Alaska increased ~40% during the 1998–2003 period while BC and US West Coast declined ~50% and 40%, respectively, during this time. The steepest declines in each of the regions were consistent in magnitude but slightly staggered in timing: Gulf of Alaska 67% (2003–2013); BC 65% (2007–2016) and US West Coast 63% (2007–2015) (Figure 1i–k). The US West Coast IPHC longline index had a similar decline over the same time-span as the trawl index even though the IPHC survey domain did not include the southern portions of the US West Coast included in the trawl surveys (IPHC survey extended into California in 2013, 2014, and 2017 only) (Figure 1k, Figure S8). Of the IPHC indices, BC had the steepest declines of all of the regions and examined time periods, closely followed by US West Coast. The Gulf of Alaska population did not decline as clearly as the other regions and comprised the majority of dogfish abundance by 2022, partially attributable to its larger surveyed area (Figure S8).

3.2 | Population Index Trends by Maturity

We categorised dogfish as immature based on a 5% maturity probability cutoff from fitted length-maturity ogives (≤65.1 cm males and ≤77.0 cm females; Figure 2a,b). All lengths are reported as total length with the tail extended. Immature dogfish were proportionally encountered more frequently in southern regions than in northern regions (Figure 2c,d). Overall, immature dogfish formed the largest maturity group by biomass and therefore can be expected to drive the overall patterns in biomass trends (Figure 3a–d). Dogfish categorised as mature females based on a 95% maturity probability (≥95.6 cm; Figure 2b,d) generally formed the smallest group by overall biomass (Figure 3a–d). Biomass trends across maturity groups were most synchronous within BC (Figure 3b) and least synchronous within the US West Coast (Figure 3c).

The steepest overall declines were for dogfish categorised as mature females in BC with a 72% decline per decade (Figure 3d,i), followed by immatures in the US West Coast (70%), maturing females in BC (58%), immatures in Gulf of Alaska (57%) and mature males in BC (56%). Considered together, mature individuals (males and females) declined the most in BC (60%), whereas US West Coast and Gulf of Alaska mature individuals were mostly stable due to the differences in trends between mature females and mature males (Figure 3e, Figure S22). Rates of population decline were similar for immature dogfish in all regions (Figure 3e,f) with 57%, 56% and 70% declines per decade, for Gulf of Alaska, BC and US West Coast, respectively. Mature females on the US West Coast and BC also had similar population index patterns since about 2007 with a spike around 2010 and steep declines of almost 94% and 60% between 2010 and 2021, respectively (Figure 3j). Note the estimate of female biomass between 2009 and 2010 in BC was three times higher from 1 year to the next. In contrast, increases were only found in mature males in Gulf of Alaska and the US West Coast, but the coastwide trend was slightly negative (Figure 3e,g).

3.3 | Spatial Distribution and Trends

By visualising spatial model predictions from the trawl-survey models for the maturity- and sex-specific groups, we can observe hotspots of dogfish biomass density (Figure 4a). The waters off the west coast of Vancouver Island and northern Washington State were hotspots of biomass density for all dogfish maturity groups other than maturing females. Waters in the southeast of the Gulf of Alaska were biomass density hotspots for all groups, with the highest biomass density in the deep waters north of Chichagof Island in Alaska (Cross Sound) (Figure S21iii). Only immature dogfish had a hotspot in the southern region of the US West Coast (off Eureka, just south of the Oregon border and north of San Francisco) (Figure S21v). There were generally lower densities of dogfish off the Aleutian Islands, northern BC, and in the waters off southern Washington State, Oregon and southern California.

Visualising coefficient estimates from spatially varying trend models illustrates areas of localised population change (Figure 4b). Across all maturity groups, there were coastwide declines with the areas of the greatest decline in Oregon/northern California, Washington/West Coast Vancouver Island and eastern Alaska. Mature females were the most universally declining group along the entire coast, with almost no areas of increasing biomass density, while immature dogfish had some of the greatest declines in biomass density (Figure 4b). Maturing males and females also showed declines across the coast; however, with fewer concentrations of steep declines than immature dogfish. In contrast, mature males had coastwide increases in biomass density with the exception of the area west of Vancouver Island, BC and northern Washington.

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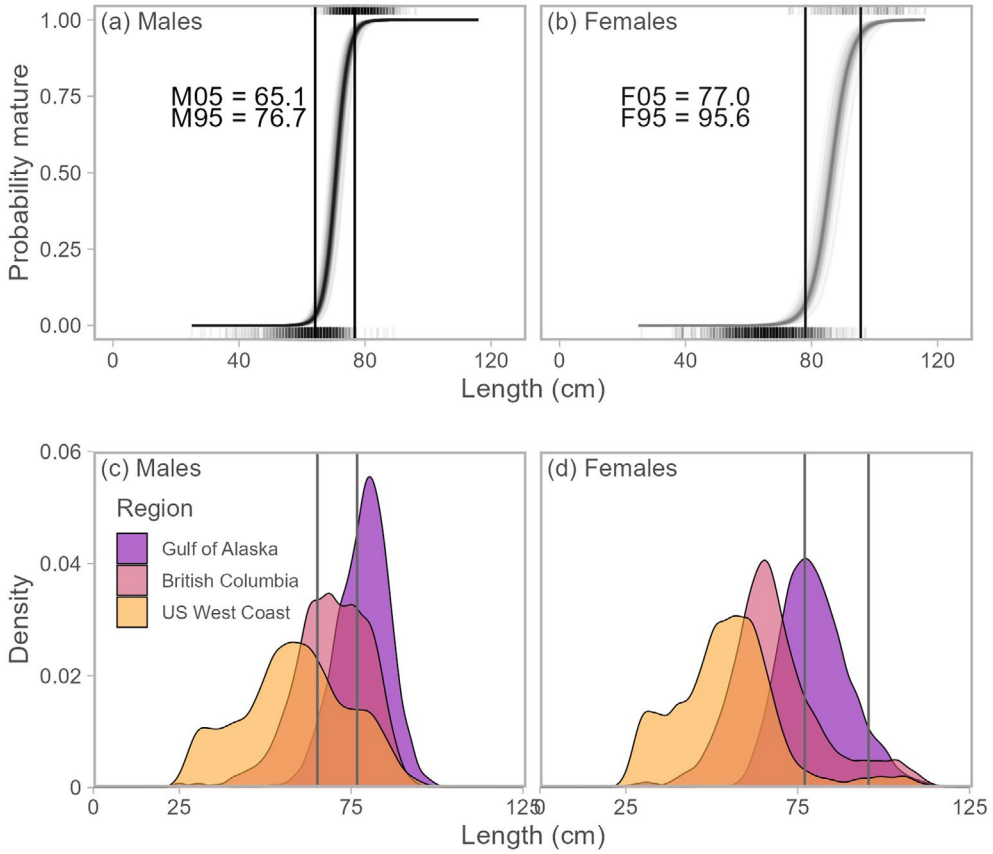


FIGURE 2 | Maturity definitions and distribution of lengths captured by trawl surveys. Maturity ogives for (a) male and (b) female dogfish with estimated lengths at maturity. Individual curved lines represent ogives by sample ID, which were treated as a random intercept in a GLMM with a logit link and Bernoulli error. Density plots of captured dogfish within regional trawl surveys for (c) males and (d) females. Vertical lines indicate lengths at 0.05 and 0.95 probability of maturity. Dogfish below the 0.05 threshold were considered 'immature'. Dogfish between 0.05 and 0.95 were considered 'maturing'. Dogfish greater than the 0.95 threshold were considered 'mature'. Length was measured with tail extended.

The waters west of Vancouver Island, BC were a consistent hotspot of estimated biomass density loss with the decline most apparent for immature dogfish, maturing males and mature males (Figure 4b). Shelikof Strait, Alaska and the waters around Kodiak Island showed one of the only consistent local increases in biomass, driven by increases in maturing females and mature males, while east of Kodiak Island there were notable declines in all stages, particularly immature and maturing stages (Figure S21ii). The US West Coast was similar with hotspots of decline except for mature males and maturing females off California. Estimates of proportional change of biomass per decade (Figure S20b) need to be interpreted in the context of local biomass density (Figure S20a). Six percent of grid cells had a decline > 80% per decade in the period of 2003–2023 for all maturity groups of dogfish (Figure S20b). Approximately 11%, 16% and 11% of grid cells in BC, Gulf of Alaska and on the US West Coast, respectively, had declines > 70% per decade after 2003 (Figure S20b).

3.4 | Trends in Depth and Temperature Occupied

We observed a decline in biomass-weighted mean depth occupied by dogfish across most maturity groups in the BC and US West Coast regions but not in the Gulf of Alaska (Figure 5). On the US West Coast, the one group that did not clearly deepen in distribution was mature females. While mature females were initially found at the deepest depths of all the maturity groups (2002) in the US West Coast, they were the shallowest maturity group in the other two regions (Figure 5). The deepening of mean depth for dogfish on the US West Coast for other maturity groups was fairly constant from about 2005 to 2017 but was more variable since then (Figure 5c). The mean depth of dogfish deepened along a north–south gradient, with stable depths in Alaska and steep depth profiles in the US West Coast, potentially suggestive of dogfish maintaining their thermal preference across latitudes (Figure 5a–f).

To understand the potential for shifts in depth to be related to shifts in response to temperature change, the plot of biomass-weighted mean temperature is best interpreted in association with the biomass-weighted depth plot (Figure 5a–f). We find that the relationship between biomass-weighted mean depth and temperature is variable across regions. While GOA dogfish depth distribution was mostly stable across maturity groups, the biomass-weighted mean temperature was highly variable and largely tracked the Alaskan temperatures had dogfish remained stationary (Figure 5a,d). In both BC and the US West Coast, dogfish maturity group density deepened over time, especially for the US West Coast populations (Figure 5b,c). While the temperature profile was somewhat consistent for BC (Figure 5e,h), potentially consistent with deepening to track preferred

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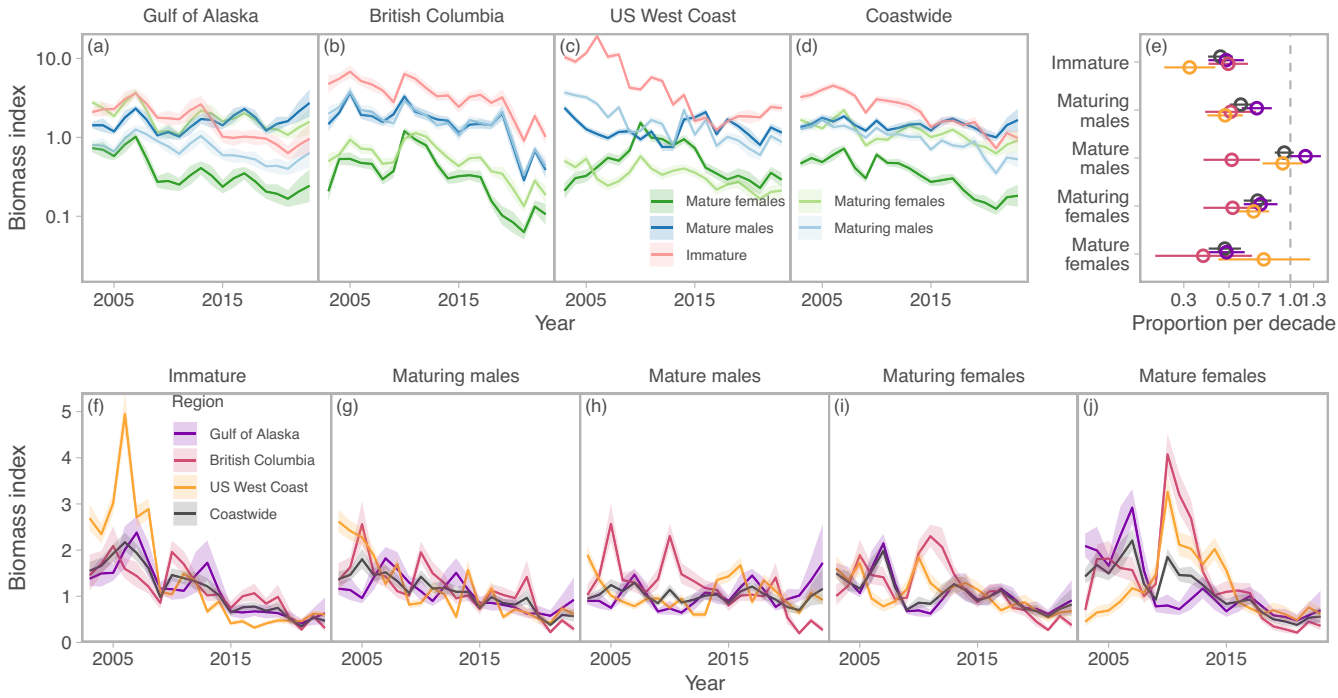


FIGURE 3 | Relative biomass trends by maturity group and region. (a–d) Biomass indices coloured by maturity group and arranged in panels by region and then all regions combined (d-coastwide). Each panel has a geometric mean of 1 and the *y* axis here is log-distributed. (e) Proportional change in biomass per decade from 2003 onwards from a log-linked GLM with gamma error. (f–j) Biomass indices coloured by region and arranged in panels by maturity group. Each time series is scaled to have a geometric mean of 1 to assess synchrony within maturity groups across regions. Lines and ribbons in panels (a–d) and (f–j) represent means and 50% confidence intervals (to minimise visual overlap), respectively. Dots and lines in panel e represent means and 95% confidence intervals.

temperatures, the biomass-weighted mean temperature in the US West Coast increased from 2010 onwards (Figure 5f)—a pattern not seen if the biomass distribution was held constant through time (Figure 5i).

4 | Discussion

Our analysis overcame several challenges with transboundary marine species population species distribution modelling and trend evaluation by combining ten fisheries-independent surveys across three regions. This enabled us to evaluate dogfish from a transboundary perspective and reconcile reports of declines and increases across regional portions of the dogfish's range. We find that dogfish have undergone substantial reductions in biomass and numbers over, at least, the past two decades. The coastwide trawl model indicated an average decline steeper than those estimated from the longline coastwide model, perhaps reflecting hyperstability due to the aggregating effect of baiting (Kuriyama et al. 2019). We found variability in the regional declines as described by the trawl survey—the US West Coast and BC had similar steep average declines in 2003–2023 (79%; 95% CI: 71%–85% and 72%; 95% CI: 58%–82%, respectively), while Alaskan declines were less steep (37%; 95% CI: 13%–54%). Although we report exponential changes in relative biomass indices for the trawl survey, we note that there are oscillations in the population in both trawl and longline indices. Dogfish categorised as immature both were the largest maturity group by biomass and had the most consistent and largest proportional change per decade in biomass across the regions.

While our analysis provided a transboundary population perspective, we note that our time series is recent and short relative to the life history and exploitation history of this species. Prior to the beginning of our coastwide time series (2003 onwards), the spawning output of dogfish was estimated to be at 10%–30% and 40%–60% of unfished spawning output in BC (DFO 2025; Anderson et al. 2025) and the US West Coast (Gertseva et al. 2021), respectively. Therefore, the population trends and declines we estimated here have occurred in a population that was already depleted.

We find that the recent declines in dogfish biomass were not obviously explained by movement alone. Determining the causal pathways of the decline requires investigation into prevailing hypotheses including climate-related shifts, overfishing, predation and changes in migration timing. Previous work found that dogfish shifted their migration timing by a month in Japan (Kanamori et al. 2023), suggesting that temporal changes in movement patterns could affect how dogfish intersect with surveys conducted at fixed times each year. A key hypothesis has been that dogfish shifted their distribution north into Alaska—potentially due to climate change (Conrath and Foy 2009; COSEWIC 2011; Orlov et al. 2012; Gertseva et al. 2021; Shiffman et al. 2022). Indeed, Alaskan (GOA) dogfish increased between 1993 and 2007 but are currently estimated to only be 20% greater than 1993 estimates following declines starting around 2003. We acknowledge that some northward movement may have occurred during 1998–2003 when BC and US West Coast populations declined while Alaska increased. Additionally, the less steep recent declines in the Gulf of Alaska compared to BC

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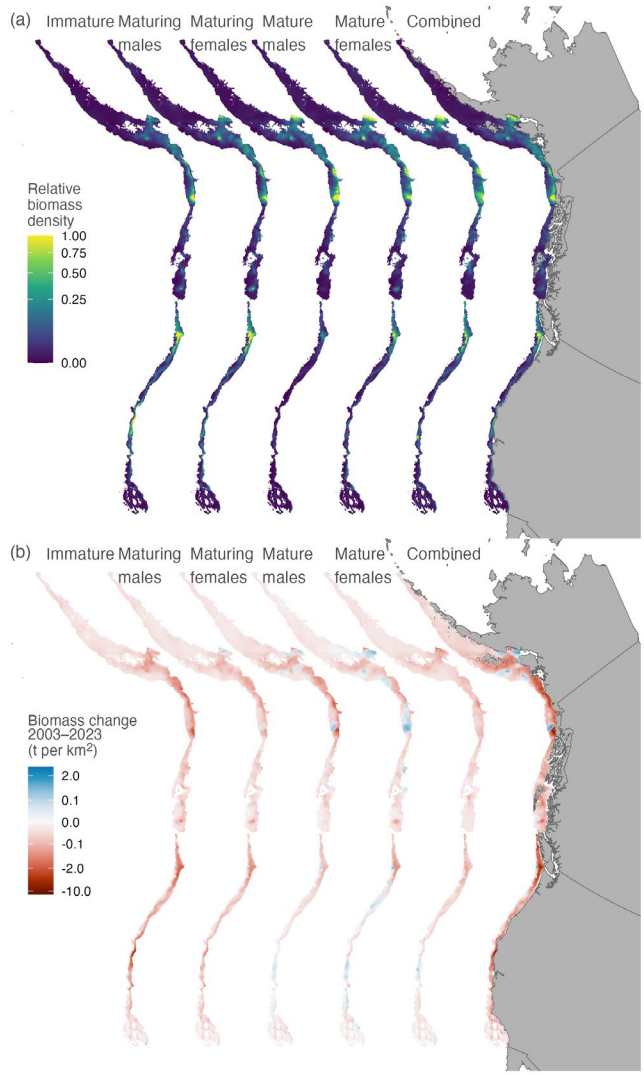


FIGURE 4 | Coastwide distribution and spatially varying relative biomass trends for dogfish for the period 2003–2023. (a) Coastwide relative biomass density in 2003 from trawl surveys. Biomass density is scaled between 0 and 1 and the colour scale is square root transformed. (b) Spatially varying biomass trends from a spatially varying coefficient model. Legend represents the difference in estimated biomass between 2003 and 2023 such that negative values reflect declines per grid cell of tonnes per km^2.

and the US West Coast could suggest continued northward redistribution. However, our results argue against redistribution as the primary explanation for coastwide declines. The coastwide biomass index declined consistently across all time periods examined, demonstrating net population loss rather than redistribution. While northward shifts may have contributed to Alaska's more gradual decline, this cannot explain the overall pattern of coastwide biomass reduction.

Additional hypotheses include changes in dogfish distribution and climate-related changes in preferred depth. We find that the biomass-weighted depth of dogfish was deepening in BC and on the US West Coast, but not in Alaska. Here, we were unable to disentangle if this deepening trend represents a shift deeper, a loss of biomass in the shallower depths, or both. If the deepening trend reflects a shift of biomass, we note that the movement to waters deeper than the survey domain is unlikely to explain the decline. While it is possible that dogfish have shifted outside of the survey domain, that is, offshore or inshore into inlets (sensu Barndoor Skate *Dipturus laevis* in the Atlantic; Simon et al. 2002; Casey and Myers 1998), additional lines of evidence suggest that this has not happened for dogfish. The trawl surveys already encompass the bulk of dogfish's depth distribution. Declines in commercial catch per unit effort from the bottom trawl fleet in BC suggest declines occurred throughout the year and across depths (Anderson et al. 2019, 2025), while a stable trend in the Alaskan deep-water longline survey (C. Tribuzio, personal communication) and in a deep-water trap survey in BC (Anderson et al. 2025) suggests dogfish have not moved to slope habitat.

Although we did not directly test environmental variables as a predictor of distribution change, our work is consistent with findings from earlier work. Localised changes to dogfish biomass density off BC were found to be negatively associated with increasing bottom temperature (English et al. 2021). Other recent work has suggested that coastwide dogfish distribution may contract in warmer years (Ward et al. 2024). Our findings are consistent with dogfish maintaining a thermal niche by deepening their depth in the more southern parts of their distribution. While GOA dogfish showed a relatively stationary distribution despite fluctuations in temperature, BC and US West Coast dogfish showed a deepening and either stable or increasing occupied temperatures. The US West Coast dogfish deepened while temperatures remained stable or increased. This potentially reflects loss of available habitat or an artefact of the loss of biomass in the northernmost parts of the US West Coast. Regardless, climate-related deepening of the population cannot sufficiently explain declines as there was no evidence of climate tracking in Alaska and dogfish are not thought to have life histories that are susceptible to climate forcing as they occupy a wide thermal breadth, are equilibrium strategists, and give birth to large live offspring (Wood et al. 1979; King and McFarlane 2003; Yatsu et al. 2008).

The main cause of shark, ray, skate and chimaera (Chondrichthyes) declines globally is overfishing (Dulvy et al. 2021). Given the slow life-history traits of sharks and rays, the signal of past fishing pressure can lag in population trends. For example, the North Atlantic Shortfin Mako (*Isurus oxyrinchus*) stock assessment found that historical fishing pressure selected immature individuals resulting in an initially stable mature population size as mature individuals were not selected. Continued fishing on immature size classes; however, resulted in growth overfishing leading to declines in biomass as older individuals died out (Winker et al. 2020, D. Courtney, personal communication). Similarly, the large sex-biased fishery on Atlantic Spiny Dogfish (*Squalus acanthias*) resulted in population trends with oscillations as the higher removals of mature female dogfish exceeded recruitment of individuals into the mature population. Consequently, the population went through periods of higher and lower recruitment (Rago and Sosebee 2013). A period of low recruitment between 1997 and 2003 was characterised by a decline in the number of pups being born and a decline in mean length of mature females (Rago and Sosebee 2013). For dogfish, past fishing effort was focused mainly on large females. The large dogfish liver oil fishery in the

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FIGURE 5 | Biomass-weighted mean depth (a–c) and mean temperature (d–f) by region and maturity group. Panels (g–i) show biomass-weighted mean temperature with dogfish distribution held constant at the average biomass per grid cell across time. The difference between rows (d–f) and rows (g–i) reflects the effect of changing dogfish distribution. Lines represent the mean and shaded ribbons represent 50% confidence intervals.

1940s peaked at ~50,000 t/year in BC and on the US West Coast and selected large females as their vitamin A potency was the highest (Barraclough 1948; King et al. 2017). This intense fishing on mature females could have driven long-term oscillations in spawning output due to a subsequent missing cohort of juvenile dogfish (Wood et al. 1979; DFO 2025; Anderson et al. 2025).

Despite lower landings of dogfish relative to historical levels in the eastern North Pacific, the potential for contemporary overfishing remains due to challenges in developing stock assessment advice, continued high incidental catches and discard mortality. Traditional stock assessments are difficult for dogfish for several reasons. For example, dogfish are difficult to age (Mcfarlane and King 2009), dogfish are a transboundary stock with the highest population density at an international border between Washington and BC with seasonal migration across the border (Gertseva et al. 2021; Taylor et al. 2013; Taylor 2008), and population indices for dogfish are not available during the early intense phases of their fishing history nearly a century ago. These issues introduce considerable challenges for regional stock assessments. Despite these challenges, a recent stock assessment in BC found that overfishing likely occurred from the 1980s until about 2010, which coincided with a decline in spawning output due to a legacy effect from the intense fishing of the 1940s (DFO 2025; Anderson et al. 2025). On the US West Coast, overfishing also occurred in the 1980s and early 2000s (Gertseva et al. 2021). While the current target reference point is associated with a spawning potential ratio of 50% for the US West Coast population, this harvest control rule is under revision as the life history of dogfish may require more conservative targets (Gertseva et al. 2021).

While dogfish are generally thought of as predators, they are also prey, and prey switching, ecosystem changes, or predator pits (where prey populations can be held at low densities due to even low amounts of predation, e.g., Swain and Benoît 2015), are all potential causes of their ongoing declines. Here, we find the steepest declines in dogfish biomass were at shallower depths and driven by declines in juvenile biomass. One possible source of predation is from Steller Sea Lions (*Eumetopias jubatus*) and California Sea Lions (*Zalophus californianus*). Across the coast, Steller Sea Lion and California Sea Lion populations have increased in recent decades with regional variation in the magnitude of the increase. In BC, the Steller Sea Lion population increased from historic lows to numbers greater than population estimates from the early 1900s and the California Sea Lion

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population has nearly tripled since 2010 (DFO 2020, 2023). On the US West Coast, California Sea Lion population numbers levelled off in 2014 after 30–40 years of increase (Lowry et al. 2022) while Steller Sea Lions have continued to increase steadily since 1975 (Laake et al. 2018). Alaskan Steller Sea Lion populations declined during the 1990s in the eastern portion of their range but between 2000 and 2004 the western population was stable and the eastern population increased (Trites et al. 2007). Additional sources of predation mortality could include Offshore Killer Whales (*Orcinus orca*), which consume sharks as part of their diet (DFO 2018). Even with constant predator populations, predation may have differential impacts on a heavily depleted versus a less-depleted population.

Many sharks and rays are threatened with extinction due to their slow life history, late age of maturity, long gestation and low fecundity (Dulvy et al. 2021). Population changes of marine species can be challenging to detect and especially so for wide-ranging, transboundary species where data may be collected by multiple surveys across multiple jurisdictions. Combining fisheries-independent survey data permitted the evaluation of transboundary population size changes while also highlighting that regional declines are not obviously due to northward movements outside of each region.

From a conservation perspective, dogfish have long been considered a nuisance species of low management and conservation concern (Mcfarlane and King 2002) and are currently not a high-valued commercial fish. Their role in the ecosystem is not well understood. They are not a high-level predator but could potentially have impacts on or influence other species via predation and nutrient transfer due to their population size and relatively ubiquitous distribution in the eastern North Pacific. Although historically abundant and once valued, evidence suggests that dogfish may now warrant elevated conservation concern, both regionally and coastwide.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Code and data to replicate the analyses in this paper are available at https://github.com/pbs-assess/dogfish-trends and are archived at https://doi.org/10.5281/zenodo.17604517.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** faf70028-sup-0001-AppendixS1.pdf.

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